

Solving Two-Step EquationsAdditional Practice 7A.01.03 **ANSWER KEY**

1. Match each property of equality to the example that shows it. Write the letter of the example on the line.

Every example below starts from $4 + 5 = 9$.

Property of Equality	Letter
Division Property of Equality	B
Subtraction Property of Equality	A
Multiplication Property of Equality	D
Symmetric Property of Equality	C
Addition Property of Equality	E

A.	$4 + 5 - 7 = 9 - 7$
B.	$(4 + 5) \div 7 = 9 \div 7$
C.	$9 = 4 + 5$
D.	$(4 + 5) \times 7 = 9 \times 7$
E.	$4 + 5 + 7 = 9 + 7$

C is the symmetric property: the two sides trade places, so $4 + 5 = 9$ becomes $9 = 4 + 5$. Nothing is added, removed or scaled. In B the divisor is 7, not 0 — the division property requires a nonzero divisor.

Use the equation $33t - 99 = 0$ for questions 2 through 5.

2. Complete the sentence to describe the equation.

The variable t is first being **multiplied** by 33, and then 99 is being **subtracted** from this product. The value of the expression $33t - 99$ is **equal** to 0.

3. Describe the operation that isolates $33t$ on the left side.

Add 99 to both sides of the equation.

Addition property of equality.

4. Describe the operation that isolates t on the left side.

Divide both sides of the equation by 33.

Division property of equality. Multiplying both sides by $1/33$ is equally correct.

5. Now solve $33t - 99 = 0$. Then check your solution by substitution.

$$33t - 99 + 99 = 0 + 99$$

$$33t = 99$$

$$33t \div 33 = 99 \div 33$$

$$t = 3$$

$$\text{Check: } 33(3) - 99 = 99 - 99 = 0 \checkmark$$

Addition property, then division property. A right side of 0 is not a special case — it behaves like any other number.

For questions 6 through 11, solve the equation and check your solution by substitution.

6. $-4 + \frac{x}{2} = 16$

$$-4 + x/2 + 4 = 16 + 4$$

$$x/2 = 20$$

$$(x/2) \times 2 = 20 \times 2$$

$$x = 40$$

$$\text{Check: } -4 + 40 \div 2 = -4 + 20 = 16 \checkmark$$

Addition property, then multiplication property.

7. $12n + 9 = 93$

$$12n + 9 - 9 = 93 - 9$$

$$12n = 84$$

$$12n \div 12 = 84 \div 12$$

$$n = 7$$

$$\text{Check: } 12(7) + 9 = 84 + 9 = 93 \checkmark$$

Subtraction property, then division property.

8. $6x - 12 = 48$

$$6x - 12 + 12 = 48 + 12$$

$$6x = 60$$

$$6x \div 6 = 60 \div 6$$

$$x = 10$$

$$\text{Check: } 6(10) - 12 = 60 - 12 = 48 \checkmark$$

Addition property, then division property.

9. $5y + 7 = 47$

$$5y + 7 - 7 = 47 - 7$$

$$5y = 40$$

$$5y \div 5 = 40 \div 5$$

$$y = 8$$

$$\text{Check: } 5(8) + 7 = 40 + 7 = 47 \checkmark$$

Subtraction property, then division property.

10. $\frac{x}{5} - 8 = 17$

$$x/5 - 8 + 8 = 17 + 8$$

$$x/5 = 25$$

$$(x/5) \times 5 = 25 \times 5$$

$$x = 125$$

$$\text{Check: } 125 \div 5 - 8 = 25 - 8 = 17 \checkmark$$

Addition property, then multiplication property.

11. $\frac{y}{5.2} + 3.4 = 4.3$

$$y/5.2 + 3.4 - 3.4 = 4.3 - 3.4$$

$$y/5.2 = 0.9$$

$$(y/5.2) \times 5.2 = 0.9 \times 5.2$$

$$y = 4.68$$

$$\text{Check: } 4.68 \div 5.2 + 3.4 = 0.9 + 3.4 = 4.3 \checkmark$$

Subtraction property, then multiplication property. A decimal in the denominator changes nothing about the method.